

StockWise

Supply Chain Consultant Review

Prepared: May 26, 2026

Classification: Internal / Confidential

EXECUTIVE SUMMARY

StockWise demonstrates a solid operational foundation: multi-location tracking, FIFO/average costing, batch and expiry management, job-order consumption, atomic database transactions, and a per-item ledger with debit/credit running balances. These are capabilities many SME systems handle poorly or not at all.

However, several structural gaps — particularly around procurement workflows, lead-time-aware reorder logic, and material planning — limit the system's ability to drive reliable supply chain decisions. This report prioritises those gaps and provides an actionable improvement roadmap.

WHAT IS WORKING WELL

- FIFO and weighted-average costing implemented correctly with atomic DB-level transactions
- Multi-location hierarchy (warehouse !' room !' shelf) with per-location stock balances
- Batch and expiry tracking with visual alerts (critical / soon / expired)
- Job order consumption — materials issued against a job with cost roll-up
- Per-item inventory ledger with debit / credit columns and running balance
- Demand forecasting baseline with configurable lookback and forecast windows
- Role-based access control (admin / manager / viewer) enforced at the API layer
- Barcode scan integrated directly into the transaction form — reduces keying errors

Ø=Ÿ4 CRITICAL — PROCESS GAPS THAT WILL CAUSE DATA PROBLEMS

1. No Purchase Order workflow

A "purchase" transaction adds stock but there is no PO !' GRN (Goods Received Note) cycle. You cannot track what was ordered versus what arrived, detect short shipments or over-deliveries, or give receiving staff a document to receive against.

Recommendation: Add a purchase_orders table with line items. A purchase transaction should reference a PO line and close it. This also gives management a committed cost figure before goods physically arrive.

2. Reorder trigger ignores supplier lead time

The current model flags an item when "days remaining" is low. The correct trigger is:

Reorder when: $\text{current_stock} < (\text{avg_daily} \times \text{lead_time_days}) + \text{safety_stock}$

An item with a 14-day supplier lead time showing "15 days remaining" looks fine — it is not.

Recommendation: Add lead_time_days to products or a suppliers table. Recalculate reorder points as $\text{avg_daily} \times (\text{lead_time} + \text{review_period})$. Add a safety stock field derived from demand variability ($< 2 \sigma F^{-1}(\frac{1-\alpha}{2})$ service-level Z-score).

3. No cycle count / physical inventory reconciliation workflow

Adjustments exist but are ad hoc. There is no structured process to schedule a count, freeze a snapshot, record physical counts, review variances, and post approved adjustments. Without this, inventory accuracy drifts silently — shrinkage and mis-picks become invisible.

Recommendation: Add a stock_counts module with the workflow: Create count !' freeze snapshot !' enter physical counts !' system shows variance by line !' manager approves !' adjustments post automatically.

Ø=ßá HIGH — MISSING CAPABILITIES THAT LIMIT USEFULNESS

4. No Bill of Materials (BOM)

Job orders consume individual items, but there is no way to define a standard material list for a job type. Every job requires manually selecting items one by one, which is slow and error-prone.

Recommendation: Add bom_templates linked to job types. When a job order is created, pre-populate the expected materials list. This also enables MRP-lite: given N open jobs, calculate what stock is required.

5. No supplier management

There are no supplier records. You cannot track who you buy each product from, their lead times, pricing history, or which supplier was used for a given purchase batch.

Recommendation: Add a suppliers table (name, contact, lead_time_days, payment_terms). Link purchase orders and inventory batches to suppliers. This also feeds directly into fixing the forecasting lead-time gap.

6. Forecasting model has no velocity classification (ABC/XYZ)

All items receive equal forecast weight regardless of consumption value or variability. ABC = value classification (A: top 70% of stock value). XYZ = demand stability (X: stable, Z: erratic). A+X items need tight safety stock; C+Z items should be managed on-demand only.

Recommendation: Add an ABC/XYZ classification page under Forecasting. Compute automatically from transaction history and use the classification to adjust reorder recommendations and safety stock targets.

7. No inventory valuation / COGS report

There is no period-end report showing: opening stock value, purchases, issues (COGS), and closing stock value. Finance cannot close their books using StockWise data without this.

Recommendation: Add a /reports/valuation page with a date-range selector. Compute the standard inventory movement statement from transactions and unit costs.

8. Job orders have no material availability check

Given current open job orders, the system cannot answer: "Do I have enough stock to complete all of them?" This forces planners to check manually or rely on memory.

Recommendation: Add a "Check Materials" action on the Job Orders page that compares BOM-required materials against current stock and flags shortages. This is the highest single-value feature for job-based businesses.

Ø=βâ MEDIUM — QUALITY AND ACCURACY IMPROVEMENTS

9. Batch traceability is one-directional

You can see what batches are in stock, but cannot answer: "Which job orders used batch LOT-2025-003?" or "Which customers received items from the recalled batch?" Critical for food, pharma, and regulated products.

Recommendation: Link inventory_transactions to inventory_batches when consuming FIFO items. Add a batch trace report: enter a batch number, see its complete journey from purchase to consumption.

10. Forecasting does not account for seasonality

A 90-day summer average will systematically under-predict peak-season demand. The model will recommend too little stock before high-demand periods.

Recommendation: Add a year-over-year comparison view — current 30-day consumption vs the same 30 days last year. Not a full seasonal model, but enough for an SME operator to see the pattern and adjust manually.

11. No stock aging / slow-mover report

Current stock is visible but not how long items have been sitting. Dead stock is invisible until a cycle count is performed. Cash tied up in ageing inventory goes undetected.

Recommendation: Add a stock aging report using inventory_batches.purchase_date. Show items with no consumption in 90 / 180 / 365 days. Flag them as slow-moving and surface their total tied-up value.

12. Transfers have no approval step

Any manager can transfer stock anywhere without review. For multi-site businesses an unauthorised transfer from a high-performing location to a slow one is a real risk.

Recommendation: Add an optional requires_approval flag per location. Transfers out of flagged locations generate a pending request that the location manager must approve before stock physically moves.

PRIORITISED IMPROVEMENT ROADMAP

Phase	Feature	Business Justification	Impact
1	Supplier management + lead times	Fixes broken reorder logic immediately	Critical
1	Purchase Order !' GRN workflow	Closes the biggest procurement process gap	Critical
2	BOM templates per job type	Enables material planning for job-based operations	High
2	Cycle counting module	Maintains inventory accuracy over time	High
2	Inventory valuation / COGS report	Required for finance to close books from system data	High
3	MRP-lite: job material availability check	Prevents job delays due to stock shortages	High
3	ABC/XYZ classification	Makes forecasting actionable, not just informational	Medium
3	Reverse batch traceability	Required for recalls and regulated products	Medium
4	Safety stock calculator	Makes reorder points statistically defensible	Medium
4	Stock aging / slow-mover report	Surfaces dead stock and unlocks tied-up cash	Medium

BO
TT
OM
LI
NE

Stoc
kWi
se
is al
read
y ah
ead
of m
ost
off-
the-
shel
f S
ME
tool
s
on c
osti
ng ri
gour

and
job-
orde

r tra
ckin
g.
The
two
gap
s
that
hurt
mo
st o
pera
tion
ally
right
no
w
are
the
abs
enc
e of
a
PO
wor
kflo
w (b
lind
rece
ivin
g)
and
the l
ead-
time
blin
d sp
ot
in fo
reca
stin
g (r
eord
ers t
rigg
er
too l
ate).
Fix t

has
e
two
first
and
the
syst
em
bec
ome
s ge
nuin
ely r
elia
ble
for p
rocu
rem
ent
deci
sion
s. E
very
thin
g el
se
in
this
repo
rt is
an e
nha
nce
men
t lay
er b
uilt
on
wha
t is
alre
ady
a str
ong
foun
dati
on.

